Math 142, F20 **Name:** Answer

Quiz 7

<u>**Instructions:**</u> **You must show all of your work to receive credits for correct answers.**

Problem 1. (5 points) Determine whether the series is convergent or divergent using the **Integral Test**.

(a) $\sum_{n=1}^{+\infty} \frac{1}{(n+1)^2}$

$a_n = \frac{1}{(n+1)^2} \Rightarrow f(x) = \frac{1}{(x+1)^2}$

$\int_1^{+\infty} f(x)dx = \int_1^{+\infty} \frac{1}{(x+1)^2} dx$

$= \left[-\frac{1}{x+1} \right]_{x=0}^{x=+\infty}$

$= \lim_{b \to +\infty} \left[-\frac{1}{b+1} - \left(-\frac{1}{2}\right) \right]$

$= 0 + \frac{1}{2} = \frac{1}{2}$

$\Rightarrow \sum_{n=1}^{+\infty} \frac{1}{(n+1)^2}$ is convergent by the Integral Test.

(b) $\sum_{n=2}^{+\infty} \frac{1}{n\sqrt{\ln n}}$

$a_n = \frac{1}{n\sqrt{\ln x}} \Rightarrow f(x) = \frac{1}{x\sqrt{\ln x}}$

$\int_2^{+\infty} f(x)dx = \int_2^{+\infty} \frac{1}{x\sqrt{\ln x}} dx$

$= \left[2\sqrt{\ln x} \right]_{x=0}^{x=+\infty}$

$= \lim_{b \to +\infty} \left[2\sqrt{\ln b} - 2\sqrt{\ln 2} \right]$

$= +\infty$

$\Rightarrow \sum_{n=2}^{+\infty} \frac{1}{n\sqrt{\ln n}}$ is divergent by the Integral Test.

Problem 2. (5 points) Determine whether the series is convergent or divergent using the **Comparison Test** or the **Limit Comparison Test**.

(a) $\sum_{n=1}^{+\infty} \frac{\cos(n)+1}{n^3}$

$a_n = \frac{\cos n + 1}{n^3}$

let us chose $b_n = \frac{2}{n^3}$ and

Since $-1 \le \cos n \le 1$, we have

$0 \le \cos n + 1 \le 2$, thus

$0 \le a_n \le b_n$ for all n.

$\sum_{n=1}^{+\infty} \frac{2}{n^3} = 2\sum_{n=1}^{+\infty} \frac{1}{n^3}$ is convergent

$\Rightarrow \sum_{n=1}^{+\infty} \frac{\cos n + 1}{n^3}$ is also convergent by the Comparison Test.

(b) $\sum_{n=1}^{+\infty} \frac{n}{\sqrt{n^3+2n+1}}$

$a_n = \frac{n}{\sqrt{n^3+2n+1}}$, let $b_n = \frac{n}{\sqrt{n^3}} = \frac{1}{\sqrt{n}}$.

$\lim_{n \to +\infty} \frac{a_n}{b_n} = \lim_{n \to +\infty} \frac{\frac{n}{\sqrt{n^3+2n+1}}}{\frac{1}{\sqrt{n}}}$

$= \lim_{n \to +\infty} \frac{n}{\sqrt{n^3+2n+1}} \cdot \frac{\sqrt{n}}{1}$

$= \lim_{n \to +\infty} \frac{\sqrt{n^3}}{\sqrt{n^3+2n+1}} = \lim_{n \to +\infty} \frac{1}{\sqrt{1+\frac{2}{n^2}+\frac{1}{n^3}}}$

$= 1 > 0$

$\sum_{n=1}^{+\infty} \frac{1}{\sqrt{n}}$ is divergent $\Rightarrow$

$\sum_{n=1}^{+\infty} \frac{n}{\sqrt{n^3+2n+1}}$ is also divergent by the Limit Comparison Test.